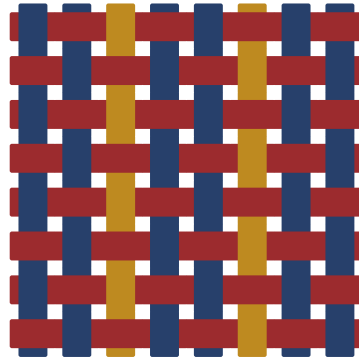




Entropy

The price of uncertainty · fill-in notes

woven on Loom

June 28, 2026

dyed in indigo, madder & weld; ruled in iron-gall. woven on Loom.

THE ONE IDEA, AND HOW TO READ THESE NOTES

□□□□

Entropy $H(X)$ is the average *surprise* of a random source, measured in bits. The single idea of these notes: entropy is a **price**. It is the exact number of bits to describe, store, or transmit X , and every theorem in the subject is one of two sentences: **you cannot pay less** (a converse) or **you needn't pay more** (an achievability).

The master table: the price, in bits (fill it in as you go)

quantity	what you are paying for
$H(X)$	bits to describe X = the best _____ rate (§3)
$H(X Y)$	bits still needed for X once you know Y
$I(X; Y)$	bits Y tells you about X = $H(X) -$ _____ (§2)
$D(p \parallel q)$	extra bits for coding p with the wrong model q ($\geq$ _____)
channel capacity C	$\max_p I(X; Y)$: bits you can send reliably (§4)

How to read (passive *and* active)

- **Read** the strands and the stated theorems.
- **Fill** the blanks _____, the [fill in: missing step] proof gaps, and the *Your turn* boxes. Any information-theory text is the answer key.
- Bump each $\backslash\mathrm{warmth}\{0\}$ to 0–5 once you can rebuild it cold.
- Logs are base 2 (bits) unless noted; base e gives nats.

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1 SURPRISE, AND ITS AVERAGE

Rare events are surprising; entropy is how surprised you are on average.

Quantify **surprise**: an event of probability p carries $-\log p$ bits of it. Certain events ($p = 1$) surprise you not at all; rare events ($p \rightarrow 0$) surprise you a lot; and surprises of *independent* events add. Entropy is just the *expected* surprise of a source: how many yes/no questions, on average, to pin down its value.

The definition

Definition 1.1 (*entropy*). For a discrete random variable X on alphabet $\mathcal{X}$ with law p ,

$$H(X) = \mathbb{E}[-\log p(X)] = - \sum_{x \in \mathcal{X}} p(x) \underline{\hspace{1cm}}.$$

It depends only on the *probabilities*, not on the values x . In bits ($\log_2$) it is the average number of fair-coin questions needed to identify X .

Your turn. Compute (in bits):

$$H(\text{fair coin}) = \underline{\hspace{1cm}}, \quad H\left(\frac{1}{4}, \frac{3}{4}\right) = \underline{\hspace{1cm}}, \quad H(\text{uniform on } 8) = \underline{\hspace{1cm}}.$$

Why $-\log p$ and not some other measure of surprise? Because it is the *only* one (up to the choice of base) that is continuous, that grows as events get rarer, and that makes the surprise of independent events **add**: $-\log(pq) = -\log p - \log q$. Additivity over independence is the hinge the whole subject swings on.

? Three sanity checks: a sure thing has $H = 0$; a fair coin has $H = 1$ bit; and H is maximized by the *uniform* law (most uncertain).
 $H\left(\frac{1}{4}, \frac{3}{4}\right) = -\frac{1}{4} \log_2 \frac{1}{4} - \frac{3}{4} \log_2 \frac{3}{4} = 1 - \frac{3}{4} \log_2 3 \approx 0.918$ bits. A biased coin is *less* uncertain than a fair one, so it is cheaper to describe.

2 THE TOOLKIT

Joint, conditional, shared, and wasted bits – all bookkeeping on one quantity.

From the single notion of entropy, four relatives organize everything: **joint** entropy (two sources together), **conditional** entropy (what's left after seeing one), **mutual information** (what they share), and **relative entropy** (the cost of a wrong model). They are tied by one picture and a couple of identities.

The four quantities

Continuous X : replace the sum by an integral to get *differential* entropy $h(X) = -\int f \log f$. It can be negative and is not coordinate-free, so handle with care; the clean, operational quantity there is relative entropy (§2).



Definition 2.1 (*the relatives*). $H(X, Y) = \mathbb{E}[-\log p(X, Y)]$, $H(X | Y) = \mathbb{E}[-\log p(X | Y)]$,

$$I(X; Y) = H(X) - H(X | Y), \quad D(p \| q) = \sum_x p(x) \log \frac{p(x)}{q(x)}.$$

The identities (the information diagram)

identity

reading

$$H(X, Y) = H(X) + H(Y | X)$$

chain rule: total = first + rest

$$I(X; Y) = H(X) + H(Y) - H(X, Y)$$

shared = overlap of the two circles

$$I(X; Y) = D(p_{XY} \| p_X p_Y)$$

information = distance from **independence**

$$I(X; Y) \geq 0, \text{ with } = 0 \text{ iff independent}$$

you never lose by looking at Y

Picture two overlapping circles, $H(X)$ and $H(Y)$. The overlap is $I(X; Y)$; the left-only crescent is $H(X | Y)$; their union is $H(X, Y)$. Every identity above is just a statement about areas in this Venn diagram, which is why information theorists draw it constantly.

Your turn. Fill the relations from the diagram:

$$I(X; Y) = H(Y) - \underline{\hspace{2cm}}.$$

$$H(X | Y) \underline{\hspace{1cm}} H(X) \quad (\text{conditioning can only } \underline{\hspace{2cm}}).$$

A caution: D is *not* a metric (not symmetric, no triangle inequality). Yet it is the right notion of "distance to the truth": the per-symbol penalty in bits for compressing data from p as if it came from q . That operational meaning is §3.

? $I(X; Y) = H(Y) - H(Y | X)$ (symmetric!); and $H(X | Y) \leq H(X)$; conditioning can only *reduce* entropy on average. Knowing more never hurts.

3 TWO PILLARS: $D \geq 0$ AND THE COMPRESSION LIMIT

□□□□ One inequality implies all the bounds; one theorem turns entropy into bits on a disk.

The whole edifice rests on two results. The first is an *inequality* from which every other bound drops out: relative entropy is never negative. The second is the *operational* payoff that makes the bits real: Shannon's source coding theorem, which says the average code length can get down to the entropy, and no further.

Pillar 1: nothing beats the truth



Theorem 3.1 (*Gibbs' inequality*). $D(p \| q) \geq 0$ for all distributions p, q , with equality iff $p = q$.

Proof. Apply Jensen to the concave log: $-D(p\|q) = \sum_x p(x) \log \frac{q(x)}{p(x)} \leq \log \sum_x p(x) \frac{q(x)}{p(x)} = \log 1 = 0$. [fill in: state the equality condition from strict concavity, and derive the two corollaries below]

Pillar 2: entropy is the floor on code length

Theorem 3.2 (*Kraft + source coding*). Any binary **prefix** code with lengths $\ell(x)$ obeys $\sum_x 2^{-\ell(x)} \leq 1$ (Kraft). Consequently the expected length $L = \sum_x p(x)\ell(x)$ satisfies

$$L \geq H(X),$$

and a code achieving $L < H(X) + 1$ always exists (take $\ell(x) = \lceil -\log p(x) \rceil$).

Proof. For the lower bound, $L - H(X) = \sum_x p(x) \log \frac{p(x)}{2^{-\ell(x)}} = D(p\|r) + \log \frac{1}{\sum 2^{-\ell}} \geq 0$, where $r(x) = 2^{-\ell(x)} / \sum 2^{-\ell}$. [fill in: check both terms are ≥ 0 using Kraft and Gibbs]

Your turn. Code the distribution $p = (\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{8})$.

$$H(X) = \text{_____ bits}, \quad L = \text{_____ bits}.$$

$$\text{optimal code lengths } (\ell_1, \ell_2, \ell_3, \ell_4) = \text{_____}.$$

Squeeze out the last bit by coding *blocks*: over n symbols the per-symbol overhead is $1/n \rightarrow 0$, so the rate $\rightarrow H(X)$. The clean statement is the AEP / typical set: about 2^{nH} sequences carry essentially all the probability, so nH bits suffice (§4).

? Two corollaries, for free, by choosing q : take q uniform to get $H(X) \leq \log |\mathcal{X}|$ (uniform is maximally uncertain); take $q = p_X p_Y$ to get $I(X; Y) \geq 0$, i.e. $H(X | Y) \leq H(X)$.

? $H = 1.75$ bits; optimal lengths (1, 2, 3, 3); $L = H$ exactly because the probabilities are **dyadic** (powers of two). Otherwise you pay up to one extra bit.

4 FACES OF ENTROPY

□□□□ Training losses, channel limits, the most honest prior: all are entropy, wearing hats.

The same price tag shows up across machine learning and communication. Each face below is entropy or one of its relatives, doing a job.

The keying

face	the idea
cross-entropy loss	training minimizes $E[-\log q_\theta] = H(p) + D(p\ q_\theta)$, i.e. it closes the KL gap to the data
channel capacity	$C = \max_p I(X; Y)$; you can communicate reliably at any rate below C , and not above
maximum entropy	the least-committed law with given constraints: fixed variance $\Rightarrow$ Gaussian, fixed mean on $\mathbb{R}_{\geq 0} \Rightarrow$ exponential
AEP / typical set	$\approx 2^{nH}$ sequences carry almost all probability, so nH bits suffice
Fano	if you must guess X from Y , your error is bounded below by $H(X Y)$

Your turn. The cross-entropy you minimize when training a classifier is $E_p[-\log q_\theta(X)]$.

- It splits as $H(p) + \underline{\hspace{2cm}}$, and only the second term depends on θ .
- So minimizing cross-entropy is exactly minimizing $\underline{\hspace{2cm}}$ from the model to the data, and the floor it can reach is $\underline{\hspace{2cm}}$.

The one idea, recovered

Read the master table again. Each row is a number of bits, and each theorem is one of two sentences. **Converses** (Gibbs, $L \geq H$, capacity, Fano) say a price cannot be undercut; **achievabilities** (Shannon codes, the AEP, capacity-achieving schemes) say it can be met. Description, compression, transmission, and learning are all the same act of paying entropy, and not a bit more than you must.

? Splits as $H(p) + D(p\|q_\theta)$; training minimizes $D(p\|q_\theta)$; the floor is $H(p)$, the data's own entropy, reached only when $q_\theta = p$.

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Where to go next: drop probability for a single string and you get **Kolmogorov complexity** (the length of the shortest program that prints it); allow lossy reconstruction and you get rate-distortion (the price of "good enough"); both are entropy's siblings.